

Foveated Shardless Worlds: A Frozen Negative Benchmark of Sparse Avatar-State Replication and Learned Local Reconstruction

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Abstract

Large shared virtual environments must decide which remote states to transmit, how often to transmit them, and what each client may safely reconstruct between authoritative updates. This paper converts a public 2022 architecture note into a bounded protocol and synthetic benchmark. The proposed composition combines distance- and view-conditioned interest management, sparse pose updates, local reconstruction, per-avatar authority, a unified coordinate frame, and explicit recovery after loss, partition, and churn. It does not claim to implement a complete serverless or shardless world.

Four systems were compared across 16, 64, and 128 simulated clients, two delivery conditions, and five seeds, yielding 120 runs. A benchmark-entry rule frozen before execution required learned reconstruction to improve visible position error or artifact rate relative to deterministic gaze / proximity reconstruction at identical transmitted bytes in at least one of six scale-condition cells, without worse recovery. The learned system passed zero cells. At 128 clients under nominal delivery, deterministic gaze / proximity routing reduced the median modeled rate from 121,920 to 12,170.65 bytes per client-second while introducing 0.0256 m mean visible position error. The learned system used the same bytes but produced catastrophic errors and a 0.6575 artifact rate.

A post hoc mathematical audit preserves that negative result while locating an implementation-specific cause. The fitted second-order velocity recurrence has spectral radius 0.99958 and remains bounded in a separate open-loop test. Failure instead appears when an authoritative observation is paired with a pre-correction prediction in the recurrence's previous-state slot. Periodic three-step resets produce an RMSE of approximately 2.91×10^{63} m/s under the frozen reset rule, compared with 0.00846 m/s when two temporally coherent observed velocities initialize the state. A new replay validator also reproduces the 120 substantive run rows, summary statistics, predictor coefficients, and failed entry decision exactly after excluding wall-clock runtime. The study therefore reports a reproducible negative benchmark, a protocol contract, and a falsifiable v2 repair hypothesis. It is not evidence that learned reconstruction is generally ineffective or that the proposed world architecture is production-ready.

Scope and evidence boundary

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1. Introduction

Imagine standing in a crowded virtual plaza. The people nearest you, the person you are looking at, and anyone interacting with you require timely updates. Avatars behind you or far away may tolerate slower updates. A client can fill short gaps by extrapolating motion, but it must not invent a purchase, a collision outcome, a permission change, or a deliberate action. When a network partition ends, provisional motion must yield to authoritative state.

This ordinary scene contains the problem studied here. A shared virtual world has limited bandwidth, changing relevance, incomplete observations, and multiple copies of remote state. Sending every state at a fixed rate is clear but scales poorly when every client tracks every other client. Sending fewer updates saves bandwidth but transfers responsibility to local reconstruction and recovery. The engineering question is therefore not simply how to compress motion. It is how to allocate updates, distinguish provisional from authoritative state, and recover safely when prediction is wrong.

Interest management, dead reckoning, and foveated streaming address parts of this problem. Visual-attention interest management can prioritize distant objects near the center of a display [2]. Goal-oriented dead reckoning allows remote replicas to act between authoritative updates while treating recovery as part of the consistency problem [3]. Gaze-driven foveated video streaming allocates visual quality nonuniformly under a bandwidth constraint [4]. These precedents establish that attention, proximity, local prediction, and selective transmission are not individually new.

The present work tests a particular composition recorded in an immutable public Git commit on June 8, 2022 [1]. The note joined:

1. temporally sparse updates whose frequency changes with virtual proximity and attention;
2. local reconstruction of omitted avatar behavior;
3. local rendering and storage rather than server-rendered graphics;
4. a narrow real-time event layer separated from historical state;
5. a unified coordinate system across instances or partitions; and
6. an eventual peer-to-peer or minimally centralized delivery architecture.

That early note was an architecture sketch, not evidence of correctness. This paper makes the testable part precise and reports the first frozen benchmark in this research program of its sparse replication and learned-reconstruction components. Three questions govern the study:

- How much bandwidth does the synthetic interest policy save relative to fixed full-state replication?
- What visible-state errors accompany those savings under nominal and adverse delivery?
- Does the tested learned recurrence improve the quality-cost frontier relative to deterministic reconstruction at the same transmitted-byte budget?

The answer to the third question is no for the frozen v1 implementation. The negative answer is preserved because it identifies a concrete state-contract failure and yields a better experimental design for a future test.

2. Prior work, contribution, and nonclaims

2.1 Prior-work boundary

The relevant earlier work supplies genuine predecessors, not obstacles to reporting a bounded new experiment.

Earlier source	Established operation	Boundary relative to this paper
Dunwell and Whelan, 2008 [2]	Display-centered visual attention refines proximity-based interest management	Does not provide per-avatar authority, learned pose reconstruction, unified cross-instance coordinates, or this message / recovery contract
Szwarcman, Feijó, and Costa, 2001 [3]	Autonomous remote replicas, goal-oriented dead reckoning, shared-state consistency, and recovery	Does not provide gaze-conditioned routing or the complete 2022 composition

Earlier source	Established operation	Boundary relative to this paper
Illahi, Siekkinen, and Masala, 2017 [4]	Eye-tracker-driven foveated video streaming under a bandwidth constraint	Adapts encoded video quality, not distributed avatar-state authority and reconstruction
Singhal and Zyda, 1999 [5]	Networked-virtual-environment design background	Does not supply this benchmark, learned recurrence, or failed entry result

The 2022 source joins sparse user-state updates, proximity and view relevance, local neural reconstruction, local rendering, unified coordinates, and a peer-to-peer or minimally centralized target [1]. The present paper does not claim that those ingredients were individually originated there. Its scientific delta is the source-bound composition made testable as a protocol, a frozen negative benchmark, an implementation diagnosis, and a substantive replay.

2.2 Contributions

This paper contributes:

- a message, authority, coordinate, consistency, and recovery contract for sparse avatar state;
- a threat model separating desired deployment properties from tested properties;
- a frozen simulator with four matched systems, three client counts, two delivery conditions, and five seeds;
- an explicit decision rule that the learned component was allowed to fail;
- a complete negative result across all six scale-condition cells;
- a post hoc recurrence and reset-state diagnosis that does not rewrite the frozen outcome; and
- an exact substantive replay check that excludes nondeterministic wall-clock measurements while preserving the original artifacts.

2.3 Nonclaims

The work does not establish:

- a complete peer-to-peer transport or discovery system;
- a shard-free persistent world database;
- cryptographic identity, privacy, Sybil resistance, or abuse resistance;
- perceptual equivalence as judged by human participants;
- production scalability or end-to-end latency on real networks;
- superiority of learned reconstruction in general;
- novelty of foveation, interest management, dead reckoning, local rendering, or learned prediction individually; or
- an exhaustive prior-art search.

The word *shardless* therefore names the target architecture, not an achieved experimental result.

3. Protocol architecture

3.1 Authority and coordinates

Each avatar has one authoritative identity key and a monotonically increasing epoch-sequence pair. A pose message is accepted only within the current epoch and only if its sequence is newer than the receiver’s stored sequence. World position is expressed using a versioned global frame identifier plus a local origin and metric transform. Crossing an instance changes routing membership, not avatar identity or world coordinates.

The separation matters because prediction is provisional. A predicted pose may be rendered locally, but it does not become authoritative history and cannot create a discrete event.

3.2 Message classes

Message	Purpose	Required semantics
HELLO_IDENTITY	Establish compatible identity and capabilities	Authenticated setup; incompatible versions rejected
STATE_KEYFRAME	Replace provisional pose with authoritative pose and velocity	Latest sequence wins within the active epoch
STATE_DELTA	Carry sparse changed fields between keyframes	Loss-tolerant; stale and duplicate sequences discarded
DISCRETE_EVENT	Represent non-predictable actions	Reliable and idempotent; never invented by the predictor
REPAIR_REQUEST	Request a new authoritative state	Triggered by age, residual, sequence gap, or reconnection
LEAVE_TOMBSTONE	Prevent stale resurrection	Epoch-bound expiry and authenticated departure

3.3 State machine and recovery

A remote avatar moves through the following local states:

ABSENT -> DISCOVERED -> TRACKING -> PREDICTING -> DEGRADED -> REPAIRING
-> TRACKING

Loss is tolerated only while an age or error budget remains valid. During a partition, cross-partition poses age toward freeze, fade, or removal. Reconnection requires a new keyframe. A teleport, epoch change, coordinate-frame change, or physically implausible residual resets the predictor. These transitions are a design contract in v1; the benchmark implements only a subset of the complete message protocol.

4. Interest and reconstruction model

4.1 Distance- and view-conditioned update intervals

For viewer i and remote avatar j , define distance

$$d_{ij}(t) = \|\mathbf{p}_j(t) - \mathbf{p}_i(t)\|_2. \quad (1)$$

The base interval in simulator steps is

$$I_{ij}^{(0)}(t) = \begin{cases} 1, & d_{ij}(t) \leq 5 \text{ m}, \\ 3, & 5 \text{ m} < d_{ij}(t) \leq 15 \text{ m}, \\ 10, & d_{ij}(t) > 15 \text{ m}. \end{cases} \quad (2)$$

Let $g_{ij}(t) = 1$ when avatar j falls within a viewer-centered 45-degree half-angle cone and $g_{ij}(t) = 0$ otherwise. In the gaze / proximity systems,

$$I_{ij}(t) = \begin{cases} I_{ij}^{(0)}(t), & g_{ij}(t) = 1, \\ \min(20, 2I_{ij}^{(0)}(t)), & g_{ij}(t) = 0. \end{cases} \quad (3)$$

The simulation uses normalized motion direction as a gaze proxy. It does not model eye tracking, head pose, covert attention, or saccades. This is an important construct-validity limitation.

4.2 Deterministic reconstruction

Between updates, the deterministic systems extrapolate the last stored velocity:

$$\hat{\mathbf{p}}_{t+1} = \hat{\mathbf{p}}_t + \Delta t \hat{\mathbf{v}}_t. \quad (4)$$

Upon delivery, position and velocity are replaced with their authoritative simulated values. This is a constant-velocity dead-reckoning baseline, not an animation or behavior model.

4.3 Learned velocity recurrence

The learned system uses ridge regression to fit a second-order vector recurrence on a separate synthetic trajectory set:

$$\hat{\mathbf{v}}_{t+1} = \hat{\mathbf{v}}_t \mathbf{A} + \hat{\mathbf{v}}_{t-1} \mathbf{B} + \mathbf{b}. \quad (5)$$

Here $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{3 \times 3}$, $\mathbf{b} \in \mathbb{R}^3$, and the ridge coefficient is 10^{-3} . The model is teacher-forced during fitting but recursively applied between received updates. It has no bounded-output transform, uncertainty estimate, maneuver class, position feature, time-since-update feature, or loss-aware state transition.

4.4 Bandwidth and visibility metrics

If message q contains m_q bytes and $u_q = 1$ when it is due for transmission, the run's modeled traffic is

$$B = \sum_q m_q u_q. \quad (6)$$

Full-state messages contain 64 modeled bytes and sparse-state messages contain 40. Headers, encryption, retransmission, discovery, relay overhead, voice, assets, and discrete-event traffic are absent. Reported bytes are therefore simulator accounting units with byte-scale interpretation, not packet captures.

A sender-receiver pair is evaluated when it lies inside the view cone or within 5 m. For the visible sample set $\mathcal{V}$, mean position error and artifact rate are

$$E_{\text{mean}} = \frac{1}{|\mathcal{V}|} \sum_{(i,j,t) \in \mathcal{V}} \|\hat{\mathbf{p}}_{ij}(t) - \mathbf{p}_j(t)\|_2, \quad (7)$$

$$R_{\text{artifact}} = \frac{1}{|\mathcal{V}|} \sum_{(i,j,t) \in \mathcal{V}} \mathbb{I}[\|\hat{\mathbf{p}}_{ij}(t) - \mathbf{p}_j(t)\|_2 > 0.3 \text{ m}]. \quad (8)$$

No human study justifies 0.3 m as a perceptual threshold. It is a frozen engineering threshold.

5. Frozen benchmark and reproducibility

5.1 Design

The benchmark crosses:

- five deterministic seeds;
- 16, 64, and 128 clients;
- nominal delivery and a combined loss-partition-churn condition; and
- four systems: fixed full-state, proximity-adaptive, gaze / proximity deterministic, and gaze / proximity learned.

This produces $5 \times 3 \times 2 \times 4 = 120$ runs. Each run contains 300 steps at 15 Hz, approximately 20 seconds of simulated time. The adverse condition includes 10% packet loss, a 40-step two-way group partition, and a 30-step churn interval affecting 10% of clients.

Synthetic trajectories combine circular motion, harmonics, randomized centers, radii, angular speeds, and phases. Because each client maintains a representation of every other client, this is a dense pairwise simulator. It is not evidence that a production overlay would or should maintain all $N(N-1)$ directed relationships. The benchmark is implemented in Python using NumPy array operations and seeded pseudorandom generation [6].

5.2 Predeclared decision rule

For each of the six client-count and delivery-condition cells, define

$$C = \mathbb{I}[B_L = B_D] \mathbb{I}[E_L < E_D \vee R_L < R_D] \mathbb{I}[T_L \leq T_D], \quad (9)$$

where L is the learned gaze / proximity system, D is the deterministic gaze / proximity system, and T is the pair of partition- and churn-recovery measures. The learned paper-entry claim required $C = 1$ in at least one cell. Every run and every failure remained reportable. This rule was frozen in the governed benchmark package before execution; the study was not registered in a public preregistration service.

5.3 Artifact custody and substantive replay

The frozen configuration, runner, result summary, and 120-run ledger have recorded SHA-256 hashes. The original result passed its implemented checks for run count, finite metrics, and complete system, condition, and scale coverage. Its configuration requested deterministic replay, but the original result did not report that check and serialized wall-clock runtime.

The correction is additive and does not modify the frozen runner or outputs. A separate validator treats the preserved run ledger as execution A, runs the same seeded simulator again as execution B, removes only `runtime_ms` and its summary quantiles, and then compares:

1. all 120 substantive run rows;
2. every substantive summary statistic;
3. predictor coefficients;
4. six learned-system decision cells; and
5. the final paper-entry decision.

All comparisons are exact in the current software environment. The replay therefore closes the previously documented substantive reproducibility gap. It does not make wall-clock duration deterministic across machines.

6. Results

6.1 Entry decision

The learned system used the same median bytes as the deterministic gaze / proximity system in all six cells. It improved neither mean visible position error nor artifact rate in any cell. Recovery was also worse in every adverse-delivery cell. The entry decision was therefore zero passes out of six.

Clients	Condition	Same bytes	Quality improvement	Recovery not worse	Cell pass
16	Nominal	Yes	No	Yes	No
16	Loss, partition, churn	Yes	No	No	No
64	Nominal	Yes	No	Yes	No
64	Loss, partition, churn	Yes	No	No	No
128	Nominal	Yes	No	Yes	No
128	Loss, partition, churn	Yes	No	No	No

6.2 Bounded bandwidth-error tradeoff

The deterministic gaze / proximity system substantially reduced modeled traffic relative to fixed full-state replication. At 128 clients under nominal delivery, median traffic fell from 121,920 to 12,170.65 bytes per client-second, a 90.0% reduction in this simulator. Mean visible position error increased from 0 to 0.0256 m, and artifact rate increased from 0 to 0.0106.

System, 128 clients	Condition	Bytes / client-second	Mean visible error (m)	Artifact rate	Partition recovery (steps)	Churn recovery (steps)
Fixed full-state	Nominal	121,920.00	0.0000	0.0000	0	0
Proximity-adaptive	Nominal	19,615.17	0.0264	0.0113	0	0
Gaze / proximity deterministic	Nominal	12,170.65	0.0256	0.0106	0	0
Gaze / proximity learned	Nominal	12,170.65	1.19×10^{37}	0.6575	0	0
Fixed full-state	Adverse	119,552.61	0.0774	0.0446	0	0
Proximity-adaptive	Adverse	18,388.86	0.1665	0.0875	0	0
Gaze / proximity deterministic	Adverse	11,365.42	0.1917	0.0897	0	0
Gaze / proximity learned	Adverse	11,365.42	1.98×10^{36}	0.7122	150	80

The deterministic result is a synthetic bandwidth-error tradeoff, not a perceptual or deployment result. View direction is only a motion-direction proxy and the byte model omits protocol overhead.

6.3 Learned reconstruction failure

The learned system’s median artifact rate ranged approximately from 0.65 to 0.73. Mean errors reached very large finite values. These values are not physically meaningful predictions; they are evidence that the recursive implementation escaped the scale of the simulated world.

The correct v1 conclusion is therefore not that a learned system almost worked. The tested learned system failed catastrophically under its actual state-update contract.

7. Post hoc mathematical diagnosis

7.1 Linear recurrence alone is not unstable

Writing Equation (5) as a first-order system gives

$$\begin{bmatrix} \hat{\mathbf{v}}_{t+1} \\ \hat{\mathbf{v}}_t \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{A}^\top & \mathbf{B}^\top \\ \mathbf{I} & \mathbf{0} \end{bmatrix}}_{\mathbf{M}} \begin{bmatrix} \hat{\mathbf{v}}_t \\ \hat{\mathbf{v}}_{t-1} \end{bmatrix} + \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}. \quad (10)$$

The fitted matrix has spectral radius

$$\rho(\mathbf{M}) = 0.999578 < 1. \quad (11)$$

Thus its homogeneous linear part is asymptotically stable, although close to marginal stability. On a separate deterministic trajectory set, recursive open-loop velocity RMSE remains below the constant-velocity baseline through the tested 64-step horizon. At one step the learned and constant-velocity errors are 0.00211 and 0.04222 m/s, respectively; at 64 steps they are 0.66885 and 1.70424 m/s. The catastrophic benchmark values cannot therefore be attributed simply to $\rho(\mathbf{M}) > 1$.

7.2 The failure occurs at the correction boundary

The v1 delivery code first computes a prediction $\hat{\mathbf{v}}_t^-$, then receives the authoritative current velocity $\mathbf{v}_t$. It initializes the next recurrence using

$$(\hat{\mathbf{v}}_t, \hat{\mathbf{v}}_{t-1}) \leftarrow (\mathbf{v}_t, \hat{\mathbf{v}}_t^-). \quad (12)$$

The second element is not the previous observed velocity. It is a pre-correction estimate for the same current time. Equation (12) therefore supplies a temporally incoherent state pair to a model trained on consecutive true velocities. Repeating this hybrid switch can be unstable even when the unforced recurrence matrix is stable.

A separate diagnostic using the frozen matrices makes the distinction visible:

Update interval	v1 pre-correction state RMSE (m/s)	Current value in both slots (m/s)	Two true sequential values (m/s)
1 step	0.0211	0.0422	0.00235
3 steps	2.91×10^{63}	0.0908	0.00846
10 steps	5.35×10^{38}	0.2545	0.0541
20 steps	4.44×10^{23}	0.4574	0.1588

This is a post hoc diagnostic, not a replacement result. It identifies a strong v2 hypothesis: the predictor state and correction operator must share an explicit temporal contract, and the correction must be bounded before the learned system is retested.

8. Threat model and deployment boundary

The architecture would expose sensitive gaze, position, body-motion, interaction, and identity data. A deployable system would require signed identity binding, replay protection, rate limits, bounded fanout, coordinate-plausibility checks, local gaze minimization, model provenance, residual-triggered repair, event-log reconciliation, tombstones, encryption at rest, and bounded retention.

None of those controls is implemented by the benchmark. Local prediction also introduces an integrity boundary: a predictor may render provisional motion, but it must never fabricate an authoritative event, purchase, permission change, collision outcome, or safety-relevant act.

9. Limitations

1. Trajectories are synthetic and comparatively smooth; human avatar behavior is more discontinuous and intentional.
2. Motion direction substitutes for gaze and cannot validate a true foveated-attention policy.
3. Five seeds support a deterministic engineering comparison, not strong population inference.
4. Quantiles are descriptive and no confidence interval or hypothesis test establishes generalization.
5. The 0.3 m artifact threshold lacks human perceptual calibration.
6. Modeled bytes exclude headers, acknowledgments, encryption, relays, retransmission, discovery, voice, assets, and discrete events.
7. Network delay is not modeled as a realistic distribution; the adverse condition combines fixed scripted loss, partition, and churn.
8. The simulator uses dense directed client pairs and does not implement a scalable overlay.
9. Runtime measurements reflect one Python environment and are not a systems-throughput benchmark.
10. The substantive replay is exact in the current environment but has not yet been reproduced on a second hardware and software stack.
11. The post hoc diagnosis explains this implementation failure but has not passed a separately frozen v2 outcome rule.
12. The prior-work review is bounded to anchor works and does not authorize a broad novelty or priority claim.

10. Predeclared v2 requirements

A v2 experiment should be created as a new immutable package rather than modifying v1. Before results are generated, it should freeze:

1. an explicit recurrence-state schema defining the timestamp of every feature;
2. a correction operator that either supplies two coherent observations or resets the recurrence to a bounded safe state;
3. output clipping or a physically justified bounded dynamics model;
4. uncertainty and time-since-update features;
5. constant-position, constant-velocity, constant-acceleration, and alpha-beta filtering baselines;
6. one-step, open-loop, periodic-reset, and packet-loss ablations;
7. substantive deterministic replay checks that exclude wall-clock fields;
8. a real packet accounting model and network-delay distribution;
9. independent methods and code review; and
10. a human perceptual study before any claim of perceptual equivalence or acceptable artifact rate.

The v2 entry rule should remain adversarial: learned reconstruction must improve a declared quality-cost frontier at equal or lower transmitted bytes without degrading repair behavior or violating bounded-state constraints.

11. Conclusion

The 2022 Foveated Shardless Worlds note proposed a coherent product direction: allocate remote-update bandwidth by proximity and view relevance, reconstruct sparse behavior locally, preserve identity across partitions, and separate provisional real-time state from durable events. The frozen v1 benchmark supports only a narrower conclusion.

Deterministic gaze / proximity routing created a large modeled bandwidth reduction with a measurable synthetic error cost. The tested learned recurrence failed all six predeclared cells and produced catastrophic errors. A mathematical audit shows that the recurrence itself is near-marginal but bounded in an isolated rollout; the catastrophic failure arises at the hybrid correction boundary, where a current observation is paired with a pre-correction estimate as if it were the previous state. A separate substantive replay reproduces the frozen result exactly after excluding wall-clock timing.

That negative result is useful. It replaces an attractive architecture claim with a precise protocol, a falsified implementation, a reproducible evidence package, and a concrete next experiment. It does not establish a shardless metaverse, but it does show how such a proposal can be tested without protecting the desired conclusion from failure.

Data and code availability

The original 2022 note is preserved in the public Self Aware Networks Git history [1]. The release package accompanying this manuscript contains the frozen v1 configuration, runner, result summary, 120-run ledger, protocol, threat model, outcome review, post hoc diagnostic, substantive replay validator, replay receipt, manuscript, and checksums. The post hoc diagnostic and replay validator are separately labeled so that neither can be mistaken for part of the predeclared v1 result.

References

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